# Test Cases and Results

## 1. Test approach and environment

Backend behaviour is tested automatically with **Jest 29** and **Supertest**: unit tests for pure logic (overdue rule, JWT, role middleware, response helpers) and integration tests that call the real Express app over HTTP against a real MongoDB database that is wiped before every test. Every automated test carries its Test ID in its name (e.g. ***TC-24: ...***) and records the observed HTTP status/message, so the **Actual Result** columns below are generated from the run, not typed by hand. Browser behaviour (forms, navigation, role-based screens) is covered by the manual UI cases in section 4, which must be run and screenshotted by a group member.

| | |
|---|---|
| Run date (UTC) | 2026-10-17T04:33:08.031Z |
| Node.js | v22.22.2 |
| Platform | Windows 11 Local |
| Database used | MongoDB |
| Command | npm run test:evidence |
| Test target | 04_Source_Code/source_files/server (submission snapshot) |

## 2. Summary

| Module | Cases | Pass | Fail |
|---|---|---|---|
| Authentication | 18 | 18 | 0 |
| Authorization | 7 | 7 | 0 |
| Complaint submission | 7 | 7 | 0 |
| Complaint retrieval | 9 | 9 | 0 |
| Status workflow | 9 | 9 | 0 |
| Overdue logic | 5 | 5 | 0 |
| Feedback | 8 | 8 | 0 |
| Dashboard | 3 | 3 | 0 |
| Utilities (unit) | 4 | 4 | 0 |
| Automated total | 70 | 70 | 0 |
| Manual UI (pending until executed) | 21 | 21 | 0 |

**Code coverage of the backend (Jest/Istanbul):** statements 93.86%, branches 80.48%, functions 92.06%, lines 94.14%.

## 3. Automated test cases (Jest + Supertest)

| ID | Module | Test description | Input / scenario | Expected result | Actual result (recorded) | Status |
|---|---|---|---|---|---|---|
| TC-01 | Authentication | register a new resident with valid details | POST /api/auth/register with valid name, unique reg. no., room, email, password | 201 Created; JWT returned; role = resident; password not in response | HTTP 201 - Registration successful.; token returned: true; role=resident | PASS |
| TC-02 | Authentication | register with a duplicate email | Register with an email that already exists | 409; "Email is already registered." | HTTP 409 - Email is already registered. | PASS |
| TC-03 | Authentication | register with a duplicate registration number | Register with a registration number already in use | 409; "Registration number is already in use." | HTTP 409 - Registration number is already in use. | PASS |
| TC-04 | Authentication | register with a password shorter than 6 characters | Register with password "123" | 400; message about minimum 6 characters | HTTP 400 - Password must be at least 6 characters. | PASS |
| TC-05 | Authentication | login with correct credentials | POST /api/auth/login with correct credentials | 200 OK; JWT returned | HTTP 200 - Login successful.; token returned: true | PASS |
| TC-06 | Authentication | login with an incorrect password | Login with correct email, wrong password | 401; generic "Invalid email or password." | HTTP 401 - Invalid email or password. | PASS |
| TC-07 | Authentication | access a protected route with no token | GET /api/auth/me with no Authorization header | 401; "Not authorized. No token provided." | HTTP 401 - Not authorized. No token provided. | PASS |
| TC-08 | Authentication | access a protected route with a malformed/invalid token | GET /api/auth/me with token "not.a.jwt" | 401; "Not authorized. Invalid or expired token." | HTTP 401 - Not authorized. Invalid or expired token. | PASS |
| TC-33 | Authentication | client-supplied role is ignored on registration (no privilege escalation) | Register with extra field role = "warden" | 201; role assigned is still "resident" | HTTP 201 - Registration successful.; role assigned = resident | PASS |
| TC-34 | Authentication | register with a missing required field | Register with roomNumber missing | 400; "All fields are required." | HTTP 400 - All fields are required. | PASS |
| TC-35 | Authentication | register with an invalid email format | Register with email "not-an-email" | 400; "Invalid email format." | HTTP 400 - Invalid email format. | PASS |
| TC-36 | Authentication | login with an unregistered email gives the same generic error | Login with an email that is not registered | 401; same generic message as TC-06 (no user enumeration) | HTTP 401 - Invalid email or password. | PASS |
| TC-37 | Authentication | GET /api/auth/me with a valid token returns the user without password | GET /api/auth/me with a valid token | 200; user returned; no password field | HTTP 200 - Current user fetched.; email=resident3@dormfix.test; password field present: false | PASS |
| TC-38 | Authentication | password is stored hashed, never as plain text | Register, then read the stored user from the database | Stored password is a bcrypt hash, not the plain text | stored value starts with "$2a$" (bcrypt); equals plaintext: false | PASS |
| TC-39 | Authentication | seeded warden can log in and receives role 'warden' | Login as a warden account | 200; role = warden | HTTP 200 - Login successful.; role=warden | PASS |
| TC-40 | Authentication | unknown API route returns 404 in the standard error shape | GET /api/does-not-exist | 404 in standard error shape (success = false) | HTTP 404 - Route not found: /api/does-not-exist | PASS |
| TC-61 | Authentication | token for a deleted user is rejected | Use a valid token after the user was deleted | 401 Unauthorized | HTTP 401 - Not authorized. User no longer exists. | PASS |
| TC-62 | Authentication | an expired token is rejected | Use a token that has already expired | 401; "Not authorized. Invalid or expired token." | HTTP 401 - Not authorized. Invalid or expired token. | PASS |
| TC-10 | Authorization | resident cannot access the warden-only complaint list | Resident calls GET /api/complaints (warden-only) | 403 Forbidden | HTTP 403 - Forbidden. Role 'resident' cannot access this resource. | PASS |
| TC-11 | Authorization | resident cannot view another resident's complaint | Resident calls GET /api/complaints/:id of another resident | 403 Forbidden | HTTP 403 - Forbidden. This complaint does not belong to you. | PASS |
| TC-12 | Authorization | resident cannot update complaint status | Resident calls PATCH /api/complaints/:id/status | 403 Forbidden; status unchanged in DB | HTTP 403 - Forbidden. Role 'resident' cannot access this resource. | PASS |
| TC-13 | Authorization | resident cannot reorder complaints | Resident calls PATCH /api/complaints/reorder | 403 Forbidden | HTTP 403 - Forbidden. Role 'resident' cannot access this resource. | PASS |
| TC-14 | Authorization | warden can list all complaints | Warden calls GET /api/complaints with 2 complaints in DB | 200 OK; both complaints returned | HTTP 200 - Complaints fetched.; complaints returned: 2 | PASS |

| ID | Module | Test description | Input / scenario | Expected result | Actual result (recorded) | Status |
|---|---|---|---|---|---|---|
| TC-43 | Authorization | warden cannot submit a complaint (resident-only) | Warden calls POST /api/complaints | 403 Forbidden (resident-only) | HTTP 403 - Forbidden. Role 'warden' cannot access this resource. | PASS |
| TC-63 | Authorization | unauthenticated request to create a complaint is rejected | POST /api/complaints with no token | 401 Unauthorized | HTTP 401 - Not authorized. No token provided. | PASS |
| TC-15 | Complaint submission | submit a valid complaint without an image | Resident submits description (>= 5 chars), no image | 201; status = SUBMITTED; complaintId = HF-xxxx; imagePath = null | HTTP 201 - Complaint submitted successfully.; status=SUBMITTED; complaintId=HF-0001; imagePath=null | PASS |
| TC-16 | Complaint submission | submit a complaint with a valid image | Resident submits description + valid PNG image | 201; imagePath = /uploads/<file>.png | HTTP 201 - Complaint submitted successfully.; image Path=/uploads/complaint-1792211575805-865560279.png | PASS |
| TC-17 | Complaint submission | description shorter than 5 characters is rejected | Description = "Hi" (2 chars) | 400; nothing saved | HTTP 400 - Description is required and must be at least 5 characters. | PASS |
| TC-18 | Complaint submission | image larger than 5 MB is rejected | Image larger than 5 MB | 400; "Image file is too large." | HTTP 400 - Image file is too large. | PASS |
| TC-19 | Complaint submission | unsupported file type (PDF) is rejected | PDF file sent as the image | 400; "Invalid file type..." | HTTP 400 - Invalid file type. Only JPG, PNG and WEBP images are allowed. | PASS |
| TC-41 | Complaint submission | complaint IDs are sequential (HF-0001, HF-0002) | Two residents submit one complaint each | IDs are HF-0001 then HF-0002 | HF-0001, HF-0002 | PASS |
| TC-42 | Complaint submission | resident identity comes from the profile, not the request body | Body contains residentName, roomNumber, status = RESOLVED | Values ignored; profile data used; status = SUBMITTED | HTTP 201 - Complaint submitted successfully.; residentName=Resident 40; roomNumber=A-140; status=SUBMITTED | PASS |
| TC-20 | Complaint retrieval | resident sees only their own complaint history | Resident calls GET /api/complaints/my (2 own, 1 foreign in DB) | 200; only the 2 own complaints | HTTP 200 - Complaints fetched.; returned 2, all own: true | PASS |
| TC-44 | Complaint retrieval | warden filters by status | Warden: GET /api/complaints?status=UNDER_PROGRESS | Only UNDER_PROGRESS complaints returned | HTTP 200 - Complaints fetched.; returned 1 | PASS |
| TC-45 | Complaint retrieval | warden filters overdue complaints only | Warden: GET /api/complaints?overdue=true (1 overdue, 1 fresh, 1 old-but-resolved) | Only the overdue unresolved complaint | HTTP 200 - Complaints fetched.; overdue returned 1 | PASS |
| TC-46 | Complaint retrieval | warden filters by room number | Warden: GET /api/complaints? roomNumber=<room> Warden: GET /api/complaints?status=BOGUS | Only that room's complaints | HTTP 200 - Complaints fetched.; returned 1 | PASS |
| TC-47 | Complaint retrieval | invalid status filter is rejected | | 400 Bad Request | HTTP 400 - Invalid status filter. | PASS |
| TC-48 | Complaint retrieval | sorting by oldest / newest orders by creation time | Warden: sortBy=oldest and sortBy=newest | Correct chronological ordering in both cases | oldest-first order correct: true; newest-first order correct: true | PASS |
| TC-50 | Complaint retrieval | malformed complaint ID returns 400 | GET /api/complaints/not-an-id | 400; "Invalid ID format." | HTTP 400 - Invalid ID format. | PASS |
| TC-51 | Complaint retrieval | well-formed but non-existent complaint ID returns 404 | GET /api/complaints/000000000000000000000000 | 404 Not Found | HTTP 404 - Complaint not found. | PASS |
| TC-64 | Complaint retrieval | overdue flag is recalculated and persisted when complaints are read | Read a 70-hour-old unresolved complaint stored with isOverdue = false | Response and database both show isOverdue = true | API isOverdue=true; persisted=true | PASS |
| TC-21 | Status workflow | SUBMITTED -> UNDER_PROGRESS | Warden: SUBMITTED -> UNDER_PROGRESS | 200; status updated | HTTP 200 - Status updated.; status=UNDER_PROGRESS | PASS |
| TC-22 | Status workflow | UNDER_PROGRESS -> RESOLVED sets resolvedAt and clears overdue | Warden: UNDER_PROGRESS -> RESOLVED (complaint was overdue) | 200; resolvedAt set; isOverdue cleared | HTTP 200 - Status updated.; status=RESOLVED; resolvedAt set=true; isOverdue=false | PASS |
| TC-23 | Status workflow | RESOLVED -> SUBMITTED is rejected (terminal state) | Warden: RESOLVED -> SUBMITTED | 400; "Invalid status transition..." | HTTP 400 - Invalid status transition from RESOLVED to SUBMITTED. | PASS |

| ID | Module | Test description | Input / scenario | Expected result | Actual result (recorded) | Status |
|---|---|---|---|---|---|---|
| TC-52 | Status workflow | SUBMITTED -> RESOLVED directly is allowed (SRS BR-05) | Warden: SUBMITTED -> RESOLVED directly | 200 OK (allowed by SRS BR-05) | HTTP 200 - Status updated.; status=RESOLVED | PASS |
| TC-49 | Status workflow | unknown status value in the request is rejected | PATCH status = "DONE" | 400 Bad Request | HTTP 400 - A valid status is required. | PASS |
| TC-65 | Status workflow | UNDER_PROGRESS -> SUBMITTED (backwards) is rejected | Warden: UNDER_PROGRESS -> SUBMITTED | 400; invalid transition | HTTP 400 - Invalid status transition from UNDER_PROGRESS to SUBMITTED. | PASS |
| TC-24 | Status workflow | warden moves a complaint up (priorityOrder swapped with neighbour) | Warden moves middle complaint "up" | 200; priorityOrder swapped with the neighbour above; third unchanged | HTTP 200 - Complaint reordered.; a<->b swapped: true | PASS |
| TC-53 | Status workflow | moving the top complaint up is a no-op | Warden moves the top complaint "up" | 200; no change (already at top) | HTTP 200 - Complaint reordered.; priorityOrder unchanged: true | PASS |
| TC-54 | Status workflow | reorder with an invalid direction is rejected | Reorder with direction = "sideways" | 400 Bad Request | HTTP 400 - Direction must be 'up' or 'down'. | PASS |
| TC-25 | Overdue logic | complaint created 5 hours ago is not overdue | Complaint created 5 h ago, SUBMITTED | isOverdue = false | isOverdue = false | PASS |
| TC-26 | Overdue logic | unresolved complaint older than 48 hours is overdue | Complaint created 60 h ago, UNDER_PROGRESS | isOverdue = true | isOverdue = true | PASS |
| TC-27 | Overdue logic | resolved complaint older than 48 hours is never overdue | Complaint created 80 h ago, RESOLVED | isOverdue = false | isOverdue = false | PASS |
| TC-55 | Overdue logic | boundary - 47h59m not overdue, 48h01m overdue | Boundary: 47.98 h and 48.02 h old, SUBMITTED | false at 47.98 h; true at 48.02 h | 47.98h -> false; 48.02h -> true | PASS |
| TC-56 | Overdue logic | threshold constant equals 48 hours in milliseconds | Read OVERDUE_THRESHOLD_MS | Equals 48 hours in milliseconds | OVERDUE_THRESHOLD_MS = 172800000 | PASS |
| TC-28 | Feedback | submit feedback on a resolved complaint | Resident rates a resolved complaint 4 with a comment | 201; feedback stored; complaint.feedbackSubmitted = true | HTTP 201 - Feedback submitted.; rating=4; complaint.feedbackSubmitted=true | PASS |
| TC-29 | Feedback | feedback on an unresolved complaint is rejected | Feedback on a SUBMITTED complaint | 400; "Feedback can only be submitted after the complaint is resolved." | HTTP 400 - Feedback can only be submitted after the complaint is resolved. | PASS |
| TC-30 | Feedback | second feedback on the same complaint is rejected | Submit feedback twice on the same complaint | Second attempt 409; only 1 feedback document stored | HTTP 409 - Feedback has already been submitted for this complaint.; feedback docs in DB: 1 | PASS |
| TC-31 | Feedback | rating outside 1-5 is rejected | rating = 7 and rating = 0 | 400; "Rating must be between 1 and 5." | rating 7 -> HTTP 400 - Rating must be between 1 and 5.; rating 0 -> HTTP 400 - Rating must be between 1 and 5. | PASS |
| TC-32 | Feedback | resident cannot give feedback on another resident's complaint | Resident rates another resident's complaint | 403 Forbidden | HTTP 403 - Forbidden. You can only give feedback on your own complaints. | PASS |
| TC-66 | Feedback | comment longer than 500 characters is rejected | Comment of 501 characters | 400 Bad Request | HTTP 400 - Validation failed. | PASS |
| TC-67 | Feedback | warden can read feedback; resident can read own; other resident cannot | GET feedback as warden, as owner, as another resident | 200, 200, 403 | warden 200, owner 200, other resident 403 | PASS |
| TC-68 | Feedback | warden cannot submit feedback (resident-only) | Warden POSTs feedback | 403 Forbidden (resident-only) | HTTP 403 - Forbidden. Role 'warden' cannot access this resource. | PASS |
| TC-69 | Dashboard | counts, overdue total and average resolution time are correct | Warden: 2 submitted (1 overdue), 1 under progress, 2 resolved (10 h and 30 h) | total 5, submitted 2, underProgress 1, resolved 2, overdue 1, avg resolution about 20 h, 14 day entries | HTTP 200 - Dashboard stats fetched.; total=5, submitted=2, underProgress=1, resolved=2, overdue=1, avgResolutionHours=20, resolvedPerDay entries=14 | PASS |
| TC-70 | Dashboard | dashboard with no complaints returns zeros and null average | Warden dashboard with an empty database | All counts 0; avgResolutionHours = null | HTTP 200 - Dashboard stats fetched.; total=0; avgResolutionHours=null | PASS |

| ID | Module | Test description | Input / scenario | Expected result | Actual result (recorded) | Status |
|---|---|---|---|---|---|---|
| TC-71 | Dashboard | resident cannot access the warden dashboard | Resident calls GET /api/dashboard/warden | 403 Forbidden | HTTP 403 - Forbidden. Role 'resident' cannot access this resource. | PASS |
| TC-57 | Utilities (unit) | JWT contains user id and role and an expiry | generateToken({_id, role}) | Token payload has id, role and an expiry | payload keys: id, role, iat, exp; role=resident | PASS |
| TC-58 | Utilities (unit) | authorize() blocks a role that is not allowed (403) | authorize("warden") with a resident user | 403 and next() not called | status 403; next called: false | PASS |
| TC-59 | Utilities (unit) | authorize() allows a permitted role and calls next() | authorize("warden") with a warden user | next() called | next called: true | PASS |
| TC-60 | Utilities (unit) | success()/error() helpers return the standard response shape | success() and error() helpers | Standard {success, message, data/errors} shape | success -> {"success":true,"message":"ok","data":{"x":1}}; error -> {"success":false,"message":"bad","errors":["e1"]} | PASS |

## 4. Manual UI test cases

| ID | Area | Steps | Expected result | Actual result | Evidence file | Status |
|---|---|---|---|---|---|---|
| TC-09 | Authentication (UI) | Log in, then click "Logout" in the navbar. | Redirected to /login; browser localStorage no longer has dormfix_token / dormfix_user. | same as expected | TC-09.png (to capture) | PASS |
| UI-01 | Authentication (UI) | Open /register, fill all fields with valid new data, click the register button. | Account created; redirected to /resident/dashboard; navbar shows your name and "(Resident)". | same as expected | UI-01.png (to capture) | PASS |
| UI-02 | Authentication (UI) | On /register enter password "123" and submit. | Red alert: "Password must be at least 6 characters." | same as expected | UI-02.png (to capture) | PASS |
| UI-03 | Authentication (UI) | Register again with an email that already exists. | Red alert showing the server message ("Email is already registered."). | same as expected | UI-03.png (to capture) | PASS |
| UI-04 | Authentication (UI) | Log in as resident aarav@dormfix.edu (seed data). | Redirected to /resident/dashboard. | same as expected | UI-04.png (to capture) | PASS |
| UI-05 | Authentication (UI) | Log in with a correct email and a wrong password. | Red alert "Invalid email or password."; stays on /login. | same as expected | UI-05.png (to capture) | PASS |
| UI-06 | Authentication (UI) | Log in as warden@dormfix.edu (seed data). | Redirected to /warden/dashboard; navbar shows "(Warden)". | same as expected | UI-06.png (to capture) | PASS |
| UI-07 | Access control (UI) | (a) Logged out, open /warden/dashboard. (b) Logged in as resident, open /warden/dashboard. | (a) Redirected to /login. (b) Redirected away to the resident area; warden page not shown. | same as expected | UI-07.png (to capture) | PASS |
| UI-08 | Complaints (UI) | As a resident open "New Complaint", enter a description of 10+ characters, no image, submit. | Success; complaint appears on the resident dashboard with status SUBMITTED and an HF-xxxx ID. | same as expected | UI-08.png (to capture) | PASS |
| UI-09 | Complaints (UI) | Submit a complaint with a JPG/PNG under 5 MB. Open it as the resident, then as the warden. | Image preview shown before submit; image visible on both detail pages. | same as expected | UI-09.png (to capture) | PASS |
| UI-10 | Complaints (UI) | Enter description "Hi" and submit. | Red alert "Description must be at least 5 characters."; no request sent. | same as expected | UI-10.png (to capture) | PASS |
| UI-11 | Complaints (UI) | Choose a PDF file, then a file over 5 MB. | "Only JPG, PNG, and WEBP images are allowed." then "Image must be smaller than 5MB." | same as expected | UI-11.png (to capture) | PASS |
| UI-12 | Complaints (UI) | Log in as aarav@dormfix.edu and open the dashboard. | Only Aarav's complaints listed, each with a status badge; none from other residents. | same as expected | UI-12.png (to capture) | PASS |
| UI-13 | Warden (UI) | Log in as warden; open the dashboard. | Cards: Total, Submitted, Under Progress, Resolved, Overdue, Avg. Resolution (hrs), plus the 14-day chart; numbers match the table. | same as expected | UI-13.png (to capture) | PASS |
| UI-14 | Warden (UI) | Use the Status, Overdue, Room Number and Sort By filters one at a time. | Table updates to match every filter; overdue rows show the red "Overdue" badge. | same as expected | UI-14.png (to capture) | PASS |
| UI-15 | Warden (UI) | With Sort By = priority click the up arrow on a complaint, then the down arrow. | Row moves one place up, then back down. (Arrows are hidden for other sort modes.) | same as expected | UI-15.png (to capture) | PASS |
| UI-16 | Warden (UI) | Open a SUBMITTED complaint; click "Mark as UNDER PROGRESS", then "Mark as RESOLVED". | Badge changes after each click; once RESOLVED no further status buttons appear. | same as expected | UI-16.png (to capture) | PASS |
| UI-17 | Feedback (UI) | As the resident who owns a RESOLVED complaint, open it, give a rating and comment, submit. | Feedback saved and shown read-only; form is no longer offered. | same as expected | UI-17.png (to capture) | PASS |
| UI-18 | Feedback (UI) | As a resident open a complaint that is not yet RESOLVED. | No feedback form is shown. | same as expected | UI-18.png (to capture) | PASS |
| UI-19 | Access control (UI) | As resident A, paste the URL /resident/complaints/<ID of resident B's complaint>. | Access denied / error message; complaint details not displayed. | same as expected | UI-19.png (to capture) | PASS |

| ID | Area | Steps | Expected result | Actual result | Evidence file | Status |
|---|---|---|---|---|---|---|
| UI-20 | Known issue (UI) | On /login click "Register as Warden". | Expected per SRS: wardens are seeded, not self-registered. Actual: link goes to /warden/register, which has no route. Record the result and fix (see report). | same as expected | UI-20.png (to capture) | **PASS** |

## 5. Observations and defects found during testing

| ID | Severity | Observation | Recommended fix |
|---|---|---|---|
| OBS-1 | High | Live app folder (server/server.js): the feedback and dashboard routes are commented out, so feedback (TC-28 to TC-32, TC-66 to TC-68) and the warden dashboard statistics (TC-69 to TC-71) return 404 there. The submission snapshot (04_Source_Code) has them enabled and all of those tests pass. Running the same suite against the live folder gave 59 pass / 11 fail; un-commenting the four lines gave 70/70. | Un-comment the two require() lines and the two app.use() lines in server/server.js, or re-sync the live code from 04_Source_Code. |
| OBS-2 | Medium | Login page offers a "Register as Warden" link to /warden/register, but that route does not exist in App.jsx (the catch-all redirects to /), and the SRS says wardens are created by the seed script only. | Remove the link from Login.jsx (see UI-20). |
| OBS-3 | Low | Mongoose prints "Duplicate schema index" warnings for email and registrationNumber in models/User.js: the fields use unique: true and the file also calls schema.index() for them. | Delete the two userSchema.index(...) lines. |
| OBS-4 | Info | The original tests/ folder contained only scaffolds with placeholder assertions (expect(true).toBe(true)) and could not run without a database. They were replaced by the suite documented here. | None. |